

パーフェクトイド塔の傾化対応について

tilt(ing)

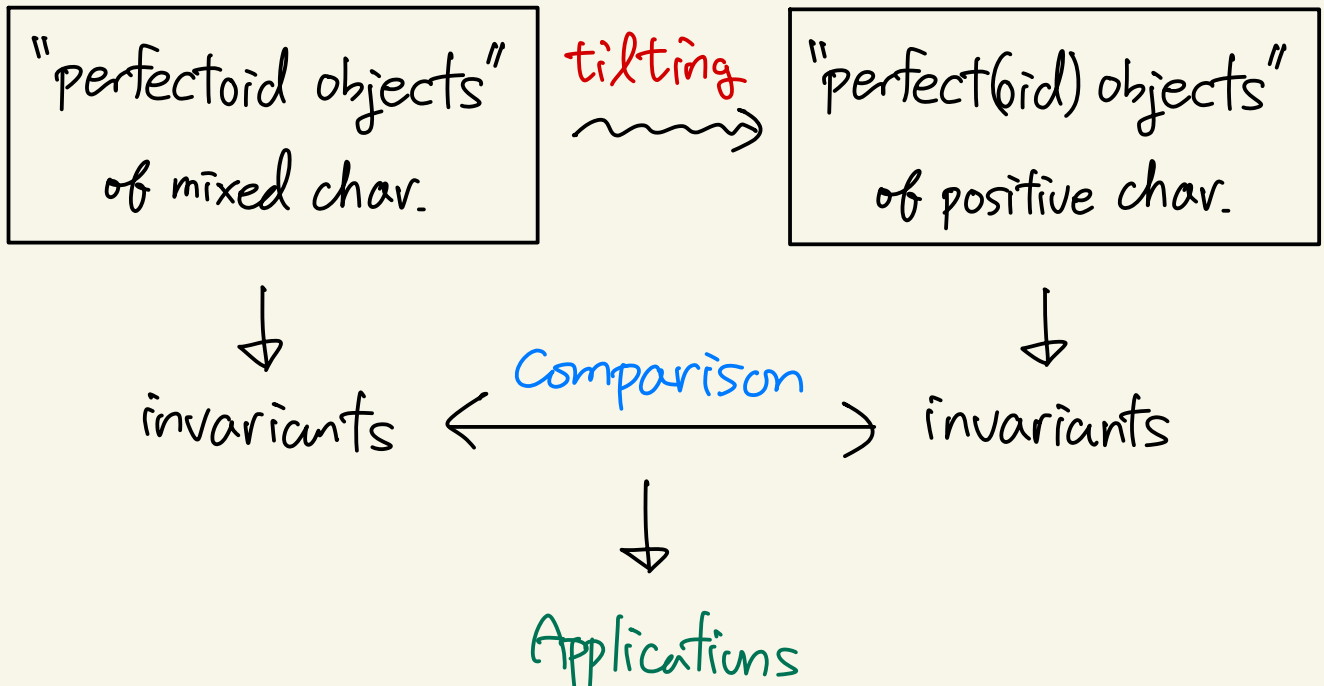
§ 1. Introduction

§ 2. Perfectoid towers

§ 3. Tilting correspondences.

§ 1

What is "tilting correspondence" ?



Perfectoid
fields

$$K = \widehat{\mathbb{Q}_p(p^{\frac{1}{p^\infty}})} \rightsquigarrow K^b = \widehat{\mathbb{F}_p((T))}(T^{\frac{1}{p^\infty}})$$
$$G_K \cong G_{K^b}$$

—//—
spaces

$$X = \mathrm{Spa}(K, \mathcal{O}_K) \rightsquigarrow X^b = \mathrm{Spa}(K^b, \mathcal{O}_{K^b})$$
$$X_{\text{ét}} \simeq X^b_{\text{ét}}$$

Applications:

p-adic Hodge theory

—//—
rings

$$A = \widehat{\mathbb{Z}_p[p^{\frac{1}{p^\infty}}]} \rightsquigarrow A^b = \widehat{\mathbb{F}_p[[T]]}[T^{\frac{1}{p^\infty}}]$$

$$\mathrm{Perfd}/_A \simeq \mathrm{Perfd}/_{A^b}$$

Direct summand conj.

BZM conj.

—//—
towers

$$\{R_i = \mathbb{Z}_p[p^{\frac{1}{p^i}}]\}_{i \geq 0} \rightsquigarrow \{R_i^{sb} = \mathbb{F}_p[[T]][T^{\frac{1}{p^i}}]\}_{i \geq 0}$$

Today

Future work.

Toy model of applications of tilting:

Prop 1.1 $A = \widehat{\mathbb{Z}_p[p^{\frac{1}{p^\infty}}]}$

$$\Rightarrow H_{\text{ét}}^i(\text{Spec } A, \mathbb{Z}/p\mathbb{Z}) = 0 \text{ for } \forall i > 1.$$

☹ We may replace A by $A^\flat = \widehat{\mathbb{F}_p[[T]]}[T^{\frac{1}{p^\infty}}]$
because

$$H_{\text{ét}}^i(\text{Spec } A, \mathbb{Z}/p\mathbb{Z}) \cong H_{\text{ét}}^i(\text{Spec } A^\flat, \mathbb{Z}/p\mathbb{Z})$$

Then we only have to use the Artin-Schreier seq.

$$0 \rightarrow \mathbb{Z}/p\mathbb{Z} \rightarrow \mathbb{G}_a \xrightarrow{F-1} \mathbb{G}_a \rightarrow 0: \text{ exact in } \widetilde{(\text{Spec } A)}_{\text{ét}}$$

↑
(small) étale topos //

§ 2

Fix a prime number p

Def 2.1 (Ishino-Nakazato-Shimamoto)

R : comm. ring

A **perfectoid tower** arising from (R, p)

is an inductive system of rings

$$\{R_i\}_{i \geq 0} : R_0 \rightarrow R_1 \rightarrow \dots \rightarrow R_i \xrightarrow{\bar{t}_i} R_{i+1} \rightarrow \dots$$

s.t. (a) $R_0 = R$

(b) $\forall i \geq 0$, $\bar{t}_i : R_i/p \rightarrow R_{i+1}/p$ is inj.

$$\begin{array}{ccc} \textcircled{c} \forall i \geq 0, & R_{i+1}/p & \xrightarrow{\text{Frob}} R_{i+1}/p \\ & \searrow \text{ } & \uparrow \bar{t}_i \\ & \exists (1) & \\ & F_i & \\ & & R_i/p \end{array}$$

i^{th} Frobenius projection

(d) $\forall i \geq 0$, F_i is surj.

(e) $\forall i \geq 0$, $p \in \text{rad}(R_i)$

(e.g., R_i is p -adically complete)

(f) $\exists \varpi \in R_1$ with $\varpi^p R_1 = p R_1$

$$\& \ker(F_i) = \varpi \cdot R_{i+1}/p R_{i+1} \quad \forall i \geq 0$$

$$\textcircled{g} \quad \forall i \geq 0, \quad P \cdot (R_i)_{P\text{-tor}} = (0),$$

$$\text{where } (R_i)_{P\text{-tor}} := \bigcup_{n \geq 0} (0 :_{R_i} P^n)$$

$$\begin{array}{ccc} (R_{i+1})_{P\text{-tor}} & \longrightarrow & R_{i+1}/P \\ \exists(1) \downarrow \scriptstyle \text{!} & \curvearrowright & \downarrow F_i \\ (R_i)_{P\text{-tor}} & \longrightarrow & R_i/P \end{array}$$

Ex 2.2

$$R_i := \mathbb{Z}_p[p^{\frac{1}{p^i}}]$$

$$\begin{array}{ccc} \mathbb{Z}_p[p^{\frac{1}{p^{i+1}}}] / P & \xrightarrow{\text{Frob}} & \mathbb{Z}_p[p^{\frac{1}{p^{i+1}}}] / P \\ \downarrow & \curvearrowright \text{---} & \uparrow \\ \mathbb{Z}_p[p^{\frac{1}{p^i}}] / P^{\frac{1}{p}} & \xrightarrow{\cong \dots} & \mathbb{Z}_p[p^{\frac{1}{p^i}}] / P \end{array}$$

Ex 2.3

R : (reduced) $\mathbb{F}_p$ -alg.

$$\begin{array}{ccc} R_{i+1} & \xrightarrow{\text{Frob}} & R_{i+1} \\ \text{Then } \cong \text{---} \uparrow & & \Leftrightarrow R_{i+1} \cong (R_i)^{\frac{1}{p}} \\ & R_i & \end{array}$$

$$\therefore \{R_i\}_{i \geq 0} \cong (R \subset R^{\frac{1}{p}} \subset R^{\frac{1}{p^2}} \subset \dots)$$

perfect tower

Def 2.4 $\{R_i\}_{i \geq 0}$: perf'd. tower arising from R .

$$R_i^{s.b} \stackrel{\text{def}}{=} \varprojlim (R_i/p \leftarrow^{F_i} R_{i+1}/p \leftarrow^{F_{i+1}} R_{i+2}/p \leftarrow \dots)$$

i^{th} small tilt

$$\begin{array}{ccc} R_{i+1}^{s.b} & \xrightarrow{F_{i+1}} & R_{i+1}^{s.b} \\ \searrow \varprojlim_j F_{j+i} & \cong & \uparrow \varprojlim_j \overline{t_{i+j}} \\ & & R_i^{s.b} \end{array}$$

$\therefore \{R_i^{s.b}\}_{i \geq 0}$ is a perfect(oid) tower.

Ex 2.5

$$R_i = \mathbb{Z}_p[p^{\frac{1}{p^i}}] \hookrightarrow \mathbb{Z}_p[T^{\frac{1}{p^i}}] / (T - p)$$

$$R_i/p \hookrightarrow \mathbb{F}_p[T^{\frac{1}{p^i}}] / (T)$$

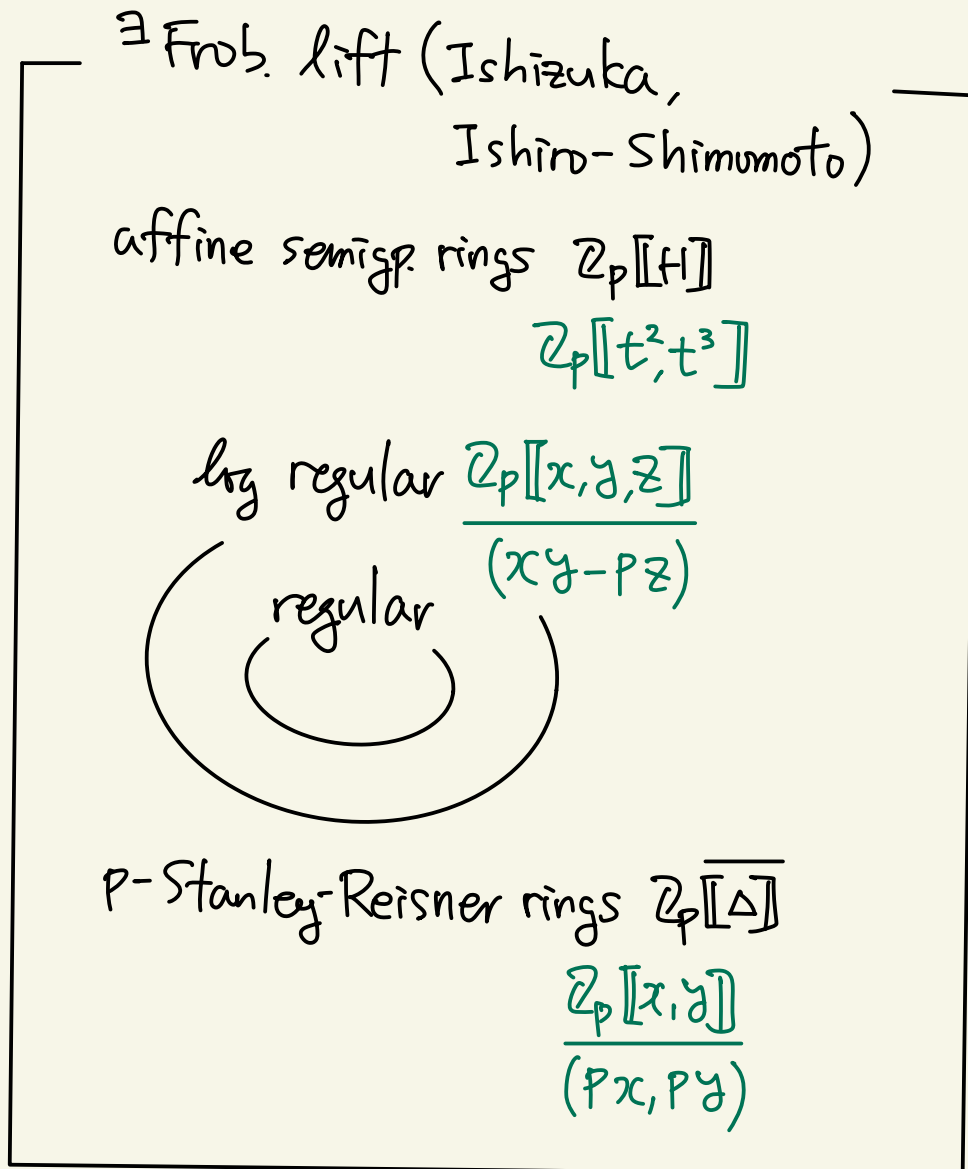
$$\mathbb{F}_p[T] / (T^{p^{i+1}}) \xrightarrow{F_{i+1}} \mathbb{F}_p[T] / (T^{p^{i+1}})$$

$$\begin{array}{ccc} & \cong & \uparrow T \mapsto TP \\ & \searrow \text{can} & \\ & & \mathbb{F}_p[T] / (T^{p^i}) \end{array}$$

$$\therefore R_i^{s.b} \cong \mathbb{F}_p[[T]]$$

$$\hookrightarrow \mathbb{F}_p[[T]]^{\frac{1}{p^i}} = \mathbb{F}_p[[T]][T^{\frac{1}{p^i}}]$$

Further constructions :



$$\mathbb{F}_p[[T]][[H]]$$

$$\mathbb{F}_p[[T]][[t^2, t^3]]$$

$$\frac{\mathbb{F}_p[[T, x, y, z]]}{(xy - Tz)}$$

$\leadsto$

$$\mathbb{F}_p[\Delta]$$

$$\frac{\mathbb{F}_p[[T, x, y]]}{(Tx, Ty)}$$

- Andreatta-Faltings' const.
(H.-Ishiro-Shimomoto)

- (Weakly) étale base change (H.)
- Modification in char. p . (H.)

$$\begin{array}{ccc}
 R & \longrightarrow & S \\
 \downarrow & \text{p.b.} & \downarrow \\
 \mathbb{F}_p \longrightarrow R' & \longrightarrow & (S/p)_{\text{red}}
 \end{array}$$

S admits a perf'd tower
 $\Rightarrow$ so does R .

§ 3. $\{R_i\}_{i \geq 0}$: a perf'd. tower arising from R .

Thm 3.1 (Ishiro - Nakazato - Shimomoto)

$$\text{Ker}(R^{s.b} \xrightarrow{p^0} R/p) = (\exists p^{s.b})$$

Moreover,

$$\textcircled{1} \quad R_i^{s.b} / p^{s.b} \xrightarrow{\cong} R_i / p$$

$$\begin{array}{ccc} \uparrow & \curvearrowright & \uparrow \end{array}$$

$$\textcircled{2} \quad (R_i^{s.b})_{p^{s.b}\text{-tur}} \xrightarrow{\cong} (R_i)_{p\text{-tur}}$$

Rem 3.2 $R_i^{s.b}$ is $p^{s.b}$ -adically complete.

Prop 3.3 $\forall i \geq 0$

① R_i is local $\Leftrightarrow R_i^{s.b}$ is local

② R_i is Noetherian $\Leftrightarrow R_i^{s.b}$ is Noetherian

③ $R_i \rightarrow R_{i+1}$ is finite $\Leftrightarrow R_i^{s.b} \rightarrow R_{i+1}^{s.b}$ is finite
 $(\Leftrightarrow R^{s.b} \text{ is } F\text{-finite})$

In ② and ③,

" $\Leftarrow$ " holds if R_i is p -adically complete.

☺ Use Thm 3.1 ① and completeness.

Say R_i is p -torsion free

$$\text{if } (R_i)_{p\text{-tor}} = 0$$

$$\Leftrightarrow p \in \text{NZD}(R_i)$$

$\downarrow$ Thm 3.1 (2),

R_i is p -torsion free $\Leftrightarrow R_i^{S.L.}$ is p -torsion free

$\therefore$ In this case, R_i is $\mathcal{CM} \Leftrightarrow R_i^{S.L.}$ is $\mathcal{CM}$.

$$\begin{pmatrix} \text{Gur.} \\ \mathcal{CI} \end{pmatrix}$$

$$\begin{pmatrix} \text{Gur.} \\ \mathcal{CI} \end{pmatrix}$$

In the general case, the following is useful.

Thm 3.5 (H.)

$$\left\{ \widehat{R}_i := R_i / (R_i)_{p\text{-tor}} \right\}_{i \geq 0}, \left\{ (R_i / pR_i)_{\text{red}} \right\}_{i \geq 0}$$

are perfect(vid) towers.

Moreover, the diagram of rings

$$\begin{array}{ccc} R_i & \longrightarrow & \widehat{R}_i \\ \downarrow & & \downarrow \\ (R_i/p)_{\text{red}} & \longrightarrow & (\widehat{R}_i/p)_{\text{red}} \end{array}$$

is cartesian.

Ex 3.6

$$\begin{array}{ccc}
 \frac{\mathbb{Z}_p[[x, y]]}{(px, py)} & \longrightarrow & \mathbb{Z}_p \\
 \downarrow & \text{p.b.} & \downarrow \\
 \mathbb{F}_p[[x, y]] & \longrightarrow & \mathbb{F}_p
 \end{array}$$

$$\begin{aligned}
 R &\cong \widetilde{R} \times \begin{matrix} (R/p)_{\text{red}} \\ (\widetilde{R}/p)_{\text{red}} \end{matrix} \\
 &\Downarrow \\
 R^{s.b} &\cong \widetilde{R^{s.b}} \times \begin{matrix} (R/p)_{\text{red}} \\ (\widetilde{R}/p)_{\text{red}} \end{matrix} \quad \widetilde{R^{s.b}} \cong (\widetilde{R})^{s.b}
 \end{aligned}$$

Cor 3.7 R_i : Noeth. local

$$\Rightarrow \dim R_i = \dim R_i^{s.b}$$

$$\odot \dim R_i = \max \left\{ \dim \widetilde{R}_i, \dim (R_i/p)_{\text{red}} \right\}$$

//

$$\dim R_i^{s.b} = \max \left\{ \dim \widetilde{R}_i^{s.b}, \dim (R_i/p)_{\text{red}} \right\} //$$

Thm 3.8 (Koszul homology)

$$\mathcal{X} = p, x_2, \dots, x_d \in R_i$$

$$\mathcal{X}^{s,b} = p^{s,b}, x_2^{s,b}, \dots, x_d^{s,b} \in R_i^{s,b}$$

: identified under Thm 3.1 ①

$$\Rightarrow K_{\bullet}(\mathcal{X}; R_i) \cong K_{\bullet}(\mathcal{X}^{s,b}; R_i^{s,b}) \text{ in } D(A_b)$$

⊙ Using exact triangles,

$$K_{\bullet}(\mathcal{X}; R_i) \rightarrow K_{\bullet}(\mathcal{X}; \widehat{R}_i) \oplus K_{\bullet}(\mathcal{X}; R_i/p) \rightarrow K_{\bullet}(\mathcal{X}; \widehat{R}_i/p) \cdots$$

$$K_{\bullet}(\mathcal{X}^{s,b}; R_i^{s,b}) \rightarrow K_{\bullet}(\mathcal{X}^{s,b}; R_i^{s,b}) \oplus K_{\bullet}(\mathcal{X}; R_i/p) \rightarrow K_{\bullet}(\mathcal{X}; \widehat{R}_i/p) \cdots$$

one can reduce to the p -torsion free case.

$$\text{Then } K_{\bullet}(\mathcal{X}; R) \xrightarrow{\cong} K_{\bullet}(\mathcal{X}'; R/p)$$

$$\uparrow \cong$$

$$K_{\bullet}(\mathcal{X}^{s,b}; R^{s,b}) \xrightarrow{\cong} K_{\bullet}(\mathcal{X}^{s,b'}; R^{s,b}/p^{s,b}) //$$

Cor 3.9 R_i : Noeth. local.

$$\Rightarrow \text{depth } R_i = \text{depth } R_i^{s,b}$$

$$\therefore R_i \text{ is CM} \Leftrightarrow R_i^{s,b} \text{ is CM.}$$

⊙ By "depth sensitivity" //

Ex 3.10 Δ : a simplicial cpx

→ Stanley-Reisner ring

$$\mathbb{Q}_p[\Delta] = \mathbb{Q}_p[x_1, \dots, x_n] / \text{(squarefree monomials)}$$

Then

$$\overline{\mathbb{Q}_p[\Delta]} := \mathbb{Q}_p[\Delta] / (x_1 - p) \text{ is } \mathbb{C}M$$

$$(\Leftrightarrow) \mathbb{F}_p[\Delta] \text{ is } \mathbb{C}M$$

↑

One can apply
Reisner's criterion.

Thm 3.11 R_i, R_{i+1} : Noeth. local

TFAE

① R_i is regular.

①^{s.b} $R_i^{s.b}$ is regular.

② $R_i \rightarrow R_{i+1}$ is flat.

②^{s.b} $R_i^{s.b} \rightarrow R_{i+1}^{s.b}$ is flat.

⊆ ② $\Leftrightarrow$ ②^{s.b} :

Use Thm 3.5 to reduce to the p-turf. case,
and apply the local criterion of flatness

①^{s.b} $\Leftrightarrow$ ②^{s.b} : By Kunz's thm.

① $\Leftrightarrow$ ② : By ② $\Leftrightarrow$ ①^{s.b}, WMA R_{i+1} : p-turf.

Then we are reduced to prove the following. //

Thm 3.12 (Gabber-Lurie)

$(R, \mathfrak{m})$: Noeth. local with $P \in \mathfrak{m}$

$\exists \pi \in \text{NZD}(R)$ s.t. $P \in \pi^P R$.

Then R is regular $\Leftrightarrow \varphi : R/\pi \rightarrow R/\pi^P$ is flat.

$$\begin{array}{ccc} \psi & & \psi \\ \mathfrak{X} & \hookrightarrow & \mathfrak{X}^P \end{array}$$